Topic 1 · Logic & Truth Tables: Worked Exam Guide

[!!!] ALWAYS QUESTION 1
PROPOSITIONAL LOGIC · 9 MARKS

4-Row Truth Table Construction, Row Deduction & Equivalence ·
CROSS-REF SPEC §1.1

WORKED EXAMPLE 1 Complete 2025 S1 Question 1 Walkthrough (9 Marks)

9 MARKS GUARANTEED

2025 S1 Exam Q1: Let p and q be primitive statements.
(a) [4 marks] Write a truth table for $q \Rightarrow (p \vee (\sim q))$. (b) [2 marks] If q is true and $q \Rightarrow (p \vee (\sim q))$ is true, what can you conclude about p? Justify by citing specific lines. (c) [3 marks] Find a logically equivalent statement with at most one connective.

ROW #	P	Q	$\sim Q$	$P \vee (\sim Q)$	$Q \Rightarrow (P \vee (\sim Q))$	REFERENCE: $Q \Rightarrow P$
Line 1	T	T	F	T ∨ F = **T**	T ⇒ T = **T** (Matches Part b!)	**T**
Line 2	T	F	T	T ∨ T = **T**	F ⇒ T = **T** (Vacuous Truth)	**T**
Line 3	F	T	F	F ∨ F = **F**	T ⇒ F = **F** (Eliminated in b!)	**F**
Line 4	F	F	T	F ∨ T = **T**	F ⇒ T = **T** (Vacuous Truth)	**T**

STEP 2 (PART B) Word-for-Word Full-Mark Justification Template

- "We are given that q is **T** and the compound statement $q \Rightarrow (p \vee (\sim q))$ is **T**.
- Looking at the truth table from Part (a), q is **T** only on **Line 1** and **Line 3**.
- However, on **Line 3** the compound statement is **F**, so Line 3 is ruled out.
- Thus only **Line 1** satisfies both premises, and on Line 1 we see that ***p* must be True (T)**."

STEP 3 (PART C) Finding the 1-Connective Equivalent Statement

The final column has truth values **(T, T, F, T)** (false only on Line 3 where q = T and p = F). Comparing this with the truth table of $q \Rightarrow p$, which also has column **(T, T, F, T)**, their columns are identical in every row. Therefore, $q \Rightarrow (p \vee (\sim q))$ is logically equivalent to $q \Rightarrow p$!

VACUOUS TRUTH GOLDEN RULE
An implication $A \Rightarrow B$ is **FALSE ONLY WHEN A = T and B = F**! Whenever the hypothesis $A = F, F \Rightarrow T$ and $F \Rightarrow F$ are BOTH **TRUE**!

1-CONNECTIVE LOOKUP FINGERPRINTS
For rows (TT, TF, FT, FF):
- $p \Rightarrow q$: **(T, F, T, T)** | $q \Rightarrow p$: **(T, T, F, T)**
- $\sim p$: **(F, F, T, T)** | $p \wedge q$: **(T, F, F, F)**

WORKED EXAMPLE 2 Formal Proof from Field Axioms of R (2024/2025 Exam Q2)

8 MARKS GUARANTEED

Exam Q2 Rule: Every single step must cite exactly one Axiom code (`A1-A5`, `M1-M5`, `D`, `O1-O2`, `OA`, `OM`). Never skip parentheses (`A3/M3`) or order swaps (`A2/M2`)!

WORKED PROOF A Multiplicative Cancellation: Assume $a \neq 0$ and $ab = ac$. Prove $b = c$.

```
Step 1: Since a ≠ 0, by M5 ∃(1/a) ∈ R\{0} with a · (1/a) = 1.
Step 2: From ab = ac (Premise), multiply both sides by (1/a):
        (1/a) · (ab) = (1/a) · (ac).
Step 3: By Axiom M3 (Associativity): ((1/a)·a)·b = ((1/a)·a)·c.
Step 4: By Axiom M2 (Commutativity): (a·(1/a))·b = (a·(1/a))·c.
Step 5: Substitute a · (1/a) = 1 (Step 1):     1 · b = 1 · c.
Step 6: By Axiom M4 (Identity 1x = x):         b = c.  [Q.E.D.]
```

WORKED PROOF B Zero Multiplication Property: Prove $0 \cdot x = 0$ for all $x \in R$.

```
Step 1: By Axiom A4 (Additive identity), 0 + 0 = 0.
Step 2: Multiply by x and apply M2:   x · (0 + 0) = x · 0.
Step 3: By Axiom D (Distributivity):  x · 0 + x · 0 = x · 0.
Step 4: By Axiom A4 on RHS:           x · 0 + x · 0 = x · 0 + 0.
Step 5: By Axiom A5, ∃-(x·0) ∈ R with (x·0) + (-(x·0)) = 0.
Step 6: Add -(x·0) to both sides and apply A3 & A5:
        x·0 + 0 = 0 ⇒ x·0 = 0 (by A4) ⇒ 0·x = 0 (by M2). [Q.E.D.]
```

WORKED PROOF C Order Axiom Inequality: If $0 < a < b$, prove $a^2 < b^2$

1. Since `0 < a` and `a < b`, by Axiom **OA** (adding `-a`) and **OM** (multiplying by positive `a > 0`): `a · a < b · a`, i.e., $a^2 < ab$.
2. Since `0 < a` and `a < b`, by Axiom **O2** (Transitivity), `0 < b`. By **OM** (multiplying `a < b` by `b > 0`): $ab < b^2$.
3. By Axiom **O2** (Transitivity) on $a^2 < ab$ and $ab < b^2$: $a^2 < b^2$!

COMMON Q2 DEDUCTION TRAP
Axiom A5 states `x + (-x) = 0` (with `-x` on the RIGHT). If your proof has `(-a) + a`, you MUST cite **Axiom A2 (Commutativity)** to flip it to `a + (-a)` before replacing it with `0`!

HOW TO NEGATE QUANTIFIERS
To negate a statement: flip every $\forall \Leftrightarrow \exists$, keep domain conditions ($\in$ **R**), and negate the predicate:
$\sim(\forall \varepsilon{>}0,\ \exists \delta{>}0 : P \Rightarrow Q) \Leftrightarrow \exists \varepsilon{>}0,\ \forall \delta{>}0 : P \wedge (\sim Q)$!

Topic 2 · Supremum & Bounds: Worked Proof Guide

[!!!] ALWAYS QUESTION 3
SUPREMUM PROOFS · 9–10 MARKS

Additive Sets sup(A+B) & Scaled Sets sup(kA) · CROSS-REF SPEC §2.1

WORKED EXAMPLE 1 Complete Proof of sup(A + B) = sup A + sup B (2025 S1 Q3)

9 MARKS EXAM Q

2025 S1 Q3: Let $A, B \subset R$ be non-empty and bounded above. Define $A + B = \{x + y : x \in A, y \in B\}$.
(a) [3 marks] Show that if α is an upper bound of A and β is an upper bound of B, then $\alpha + \beta$ is an upper bound of $A + B$.
(b) [6 marks] Show that $\sup(A + B) = \sup A + \sup B$.

PART (a) PROOF **Prove $\alpha + \beta$ is an Upper Bound of $A + B$ [3 Marks]**

1. Let $z \in A + B$ be arbitrary. By definition of $A + B$, there exist $x \in A$ and $y \in B$ such that $z = x + y$.
2. Since α is an upper bound of A, we have $x \leq \alpha$. Since β is an upper bound of B, we have $y \leq \beta$.
3. Adding the two inequalities yields $z = x + y \leq \alpha + \beta$. Since this holds for all $z \in A + B$, $\alpha + \beta$ is an upper bound of $A + B$. ▪

PART (b) PROOF **Two-Way Inequality Method OR ε-Approximation Method [6 Marks]**

- **Direction 1 ($\sup(A + B) \leq \sup A + \sup B$):** Since $\sup A$ is an upper bound of A and $\sup B$ is an upper bound of B, by Part (a), $\sup A + \sup B$ is an upper bound of $A + B$. Since $\sup(A + B)$ is the *least* upper bound of $A + B$, we immediately have $\sup(A + B) \leq \sup A + \sup B$.
- **Direction 2 ($\sup(A + B) \geq \sup A + \sup B$ — Method A: ε-Doorstep Trick):** Let $\varepsilon > 0$ be arbitrary. Since $\sup A - \varepsilon/2 < \sup A$, it is not an upper bound of A, so **there exists** $x \in A$ with $x > \sup A - \varepsilon/2$. Similarly, **there exists** $y \in B$ with $y > \sup B - \varepsilon/2$.
Then $x + y \in A + B$, so $\sup(A + B) \geq x + y > (\sup A + \sup B) - \varepsilon$. Since this holds for all $\varepsilon > 0$, $\sup(A + B) \geq \sup A + \sup B$. Combining both directions gives equality! ▪

ALTERNATIVE **Scaled Set Proof: $\sup(kA) = k \sup A$ for $k > 0$**

1. By Part (a) with $\alpha = \sup A$, $k \sup A$ is an upper bound of kA, so $\sup(kA) \leq k \sup A$.
2. Conversely, since $\sup(kA)$ is an upper bound of kA, for all $x \in A$ we have $kx \leq \sup(kA) \Rightarrow x \leq \sup(kA)/k$. Thus $\sup(kA)/k$ is an upper bound of A, so $\sup A \leq \sup(kA)/k \Rightarrow k \sup A \leq \sup(kA)$!

QUANTIFIER TRAP IN SUP PROOF
When dropping below $\sup A$ to $\sup A - \varepsilon$, NEVER write "for all $x \in A$"! Only **at least one element** pierces above $\sup A - \varepsilon$: you MUST write "**there exists** $x \in A$" ($\exists x \in A$)!

WHAT IF K < 0? (NEGATIVE SCALE)
Multiplying an inequality by $k < 0$ **flips the inequality sign**, turning upper bounds into lower bounds: for $k < 0$, $\sup(kA) = k \cdot \inf A$ and $\inf(kA) = k \cdot \sup A$!

WORKED EXAMPLE 2 Constructing Specific Examples & Limit Points (Exam Q4 — 12 Marks)

12 MARKS GUARANTEED

Exam Q4 Pattern: "For each situation, provide a **specific example** (explicit formula/set) and **briefly justify** your answer." (3 parts × 4 marks = 12 marks).

CASE 1 (LIMIT POINTS) **Infinite subset $E \subset R$ whose set of limit points is $\{2\}$**

- **Specific Example:** Let $E = \{ 2 + 1/n : n \in N+ \}$.
- **Justification:** E is an infinite subset of R. Since $\lim_{n\to\infty} (2 + 1/n) = 2$ and every element $2 + 1/n \neq 2$, every open neighbourhood $(2 - \varepsilon, 2 + \varepsilon)$ contains infinitely many points of E, so 2 is a limit point. For any $L \neq 2$, choosing $\varepsilon = |L - 2|/2$ isolates L from almost all terms, so no other limit point exists. Thus the set of limit points is $\{2\}$.

CASE 2 (SEQUENCES) **Sequence (a_n) where $(a_n)^2 \to 16$, but (a_n) diverges**

- **Specific Example:** Let $a_n = 4 \cdot (-1)^n$ for each $n \in N+$.
- **Justification:** For every $n \in N+$, $(a_n)^2 = 16 \cdot ((-1)^n)^2 = 16$, which is a constant sequence converging to 16. However, (a_n) alternates between -4 (odd n) and $+4$ (even n); since two subsequences converge to different limits ($-4 \neq 4$), (a_n) diverges.

CASE 3 (CANCELLATION) **Two divergent sequences (x_n), (y_n) whose sum $(x_n + y_n) \to 3$**

- **Specific Example:** Let $x_n = 3 + (-1)^n$ and $y_n = -(-1)^n = (-1)^{n+1}$.
- **Justification:** Both (x_n) (oscillating between 2 and 4) and (y_n) (oscillating between 1 and -1) diverge. However, for all $n \in N+$, their sum is the constant sequence $x_n + y_n = 3 + (-1)^n - (-1)^n = 3$, which converges to 3!

BONUS Q4 EXAM PROMPTS	EXACT EXAMPLE TO WRITE DOWN
Set E with **two** limit points $\{0, 5\}$	$E = \{ 1/n : n \in N+ \} \cup \{ 5 + 1/n : n \in N+ \}$
Continuous f on $(0, 1)$ that is **not bounded**	$f(x) = 1 / x$ on open interval $(0, 1)$ (EVT fails on open intervals!)
Continuous f at $x = 0$ that is **not differentiable** at 0	$f(x) = \|x\|$ (Left derivative = $-1 \neq$ Right derivative = $+1$ at $x = 0$)

NEVER DRAW PICTURES ONLY IN Q4
In Q4, markers require an **explicit mathematical formula** (e.g. $a_n = 4(-1)^n$). Vague prose like "a sequence that jumps back and forth" scores 0/4!

LIMIT POINT VS SUPREMUM
A supremum does NOT have to be a limit point! For $S = \{1/n\} \cup \{5\}$, $\sup S = 5$ (isolated point), while the only limit point is 0!

Topic 3 · Sequences & MCT: Worked Proof Guide

Formal ε-N Proofs & 13-Mark Recursive Sequence Blueprint ·

CROSS-REF SPEC §3.1-3.2

[!!!] ALWAYS QUESTION 5
RECURSIVE MCT · 13 MARKS

WORKED EXAMPLE 1 Formal ε-N Sequence Limit Proof (Scratch Work → Formal Proof)

8 MARKS CORE SKILL

Exam Problem: Using the formal ε-N definition of a limit, prove that $\lim_{n\to\infty} (2n + 1) / (n + 3) = 2$.

STAGE 1 (DRAFT PAPER) Backwards Scratch Work to Bound $|x_n - L|$ by a Simple C / n

1. **Combine Fraction:** $|(2n + 1)/(n + 3) - 2| = |(2n + 1 - 2n - 6)/(n + 3)| = |-5 / (n + 3)| = 5 / (n + 3)$.
2. **Shrink Denominator to Make Fraction Larger:** Since $n + 3 > n > 0$, dropping $+3$ gives $5 / (n + 3) < 5 / n$.
3. **Solve for *n*:** We force $5 / n < \varepsilon \Leftrightarrow n > 5 / \varepsilon$. Thus our threshold is $N = \lfloor 5 / \varepsilon \rfloor$ (or any integer $N > 5/\varepsilon$)!

STAGE 2 (EXAM SCRIPT) Word-for-Word Formal Proof Write-Up (Copy This Structure!)

```
Proof: Let ε > 0 be given. By the Archimedean Property, choose N ∈ N+
such that N > 5 / ε (which implies 5 / N < ε).
Then for all integers n > N, we have:
    | (2n + 1)/(n + 3) - 2 | = | (2n + 1 - 2(n + 3)) / (n + 3) |
                             = 5 / (n + 3)
                             < 5 / n          (since n + 3 > n > 0)
                             < 5 / N          (since n > N > 0)
                             < 5 / (5 / ε) = ε.
Therefore, by the formal definition, lim_{n→∞} (2n + 1)/(n + 3) = 2.  ▪
```

DENOMINATOR TRAP What if Denominator Has a Minus Sign? e.g. $(3n + 1) / (2n - 5)$

You CANNOT write $1 / (2n - 5) < 1 / (2n)$ because subtracting 5 makes the denominator *smaller* (fraction larger)!
Fix: Restrict $n \geq 5$ so that $2n - 5 \geq n$, and then choose $N = \max\{5, \lceil C / \varepsilon \rceil\}$!

NEVER SOLVE FOR N IN FORMAL PROOF
In the formal write-up, NEVER start with $|x_n - L| < \varepsilon$ and work backwards! You MUST state $\text{Let } \varepsilon > 0$, define N first, and chain $|x_n - L| = \ldots < \varepsilon$ forward!

CONVERGENT ⇔ CAUCHY IN R
In the real numbers *R*, a sequence (x_n) is a **Cauchy sequence if and only if it converges**! Once MCT proves convergence, Cauchy is automatic!

WORKED EXAMPLE 2 Recursive Sequence, Induction, MCT & Cauchy (2025 S1 Q5 — 13 Marks!)

13 MARKS EXAM Q

2025 S1 Q5: Let $x_1 = 0$ and $x_{n+1} = 9 / (6 - x_n)$ for $n \in N+$. (a) [5 marks] Prove by induction that $0 \leq x_n < 3$ for all $n \in N+$. (b) [5 marks] Is (x_n) monotone increasing, decreasing, or neither? (c) [3 marks] Is (x_n) Cauchy? Find $\lim_{n\to\infty} x_n$.

PART (a) [5 MARKS] Mathematical Induction for Boundedness $0 \leq x_n < 3$

- **Base Case ($n = 1$):** $x_1 = 0$, which satisfies $0 \leq 0 < 3$. Thus the statement holds for $n = 1$.
- **Inductive Hypothesis:** Assume $0 \leq x_k < 3$ holds for some integer $k \in N+$.
- **Inductive Step ($n = k + 1$):** From $0 \leq x_k < 3$, negate to get $-3 < -x_k \leq 0$. Add 6 to get $3 < 6 - x_k \leq 6$.

Taking reciprocals reverses inequalities: $1/6 \leq 1 / (6 - x_k) < 1/3$.
Multiplying by 9 gives:
$9/6 \leq 9 / (6 - x_k) < 9/3 \Rightarrow 0 \leq 1.5 \leq x_{k+1} < 3$. Thus the statement holds for $n = k + 1$.

- **Conclusion:** By the Principle of Mathematical Induction, $0 \leq x_n < 3$ for all $n \in N+$. ▪

PART (b) [5 MARKS] Monotonicity via Algebraic Difference $x_{n+1} - x_n$

Compute the difference and put over a common denominator:
$x_{n+1} - x_n = 9 / (6 - x_n) - x_n = [\, 9 - x_n(6 - x_n) \,] / (6 - x_n) = (x_n^2 - 6x_n + 9) / (6 - x_n) = (x_n - 3)^2 / (6 - x_n)$.
From Part (a), $0 \leq x_n < 3$ for all $n \in N+$:

- Numerator: Since $x_n < 3$ (so $x_n \neq 3$), $(x_n - 3)^2 > 0$.
- Denominator: Since $x_n < 3$, $6 - x_n > 6 - 3 = 3 > 0$.

Thus $x_{n+1} - x_n > 0$ for all $n \in N+$, so (x_n) is **strictly monotone increasing**! ▪

PART (c) [3 MARKS] MCT, Cauchy Justification & Fixed-Point Limit Calculation

1. Since (x_n) is monotone increasing (Part b) and bounded above by 3 (Part a), by the **Monotone Convergence Theorem (MCT)**, (x_n) converges to some real limit $L \in R$.
2. Since every convergent sequence in *R* is Cauchy, **YES, (x_n) is a Cauchy sequence**.
3. Taking $n \to \infty$ on both sides of $x_{n+1} = 9 / (6 - x_n)$ yields $L = 9 / (6 - L) \Rightarrow 6L - L^2 = 9 \Rightarrow (L - 3)^2 = 0 \Rightarrow L = 3$. ▪

MANDATORY PMI CLOSING SENTENCE
In Part (a), you MUST write the final concluding sentence citing **"By the Principle of Mathematical Induction..."** or Melbourne Uni markers deduct 1 mark!

TWO ROOTS IN QUADRATIC LIMIT?
If solving $L = g(L)$ gives two roots (e.g. $L = 1$ and $L = 4$), use your Part (a) bound (e.g. $0 \leq x_n \leq 2 \Rightarrow 0 \leq L \leq 2$) to reject the extraneous root!

Topic 4 · Function Limits & Discontinuity (Q6)

[!!!] ALWAYS QUESTION 6
LIMITS & CONTINUITY · 11 MARKS

Formal ε-M Limit Proof & ε-δ Discontinuity Negation · CROSS-REF SPEC §4.1

WORKED EXAMPLE 1 Piecewise ε-M Limit & ε-δ Discontinuity (2025 S1 Q6 — 11 Marks)

11 MARKS EXAM Q

2025 S1 Q6 Archetype: Let $f : R \to R$ be defined by $f(x) = 3x - 5$ if $x < 1$, and $f(x) = 2x / (x + 1)$ if $x \geq 1$.
(a) [6 marks] Using the ε-M definition of limits, show that $\lim_{x\to\infty} f(x) = 2$.
(b) [5 marks] Using the ε-δ definition of continuity, show that f is **not continuous** at $x = 1$.

PART (a) [6 MARKS] Formal ε-M Proof for $\lim_{x\to\infty} f(x) = 2$

• **Scratch Work:** For $x \geq 1$, $f(x) = 2x/(x+1)$. Distance: $|2x/(x+1) - 2| = |-2/(x+1)| = 2/(x+1) < 2/x$. To make $2/x < \varepsilon$, we need $x > 2/\varepsilon$. To also stay in the $x \geq 1$ branch, pick $M = \max\{1, 2/\varepsilon\}$!
• **Formal Proof Write-Up:**
"Let $\varepsilon > 0$ be given. Choose $M = \max\{1, 2/\varepsilon\}$. Then for all $x > M$, since $x > M \geq 1$, we have $f(x) = 2x/(x+1)$ and $x > 2/\varepsilon > 0$. Therefore:
$|f(x) - 2| = |2x/(x+1) - 2| = 2 / (x + 1) < 2 / x < 2 / M \leq 2 / (2/\varepsilon) = \varepsilon$.
Hence, by the definition of the limit at infinity, $\lim_{x\to\infty} f(x) = 2$." ▪

PART (b) [5 MARKS] Discontinuity Proof at Jump Point $x = 1$ (Negation of ε-δ)

• **Step 1 (Compute Jump Gap):** At $a = 1$, $f(1) = 2(1)/(1+1) = 1$. As $x \to 1^-$, $3x - 5 \to -2$. Jump gap = $|1 - (-2)| = 3$. Pick $\varepsilon_0 = 1$ (any positive number ≤ 3 works!).
• **Step 2 (Formal Proof Write-Up):**
"To prove f is discontinuous at $a = 1$, we show: $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0$, $\exists x \in R$ with $|x - 1| < \delta$ and $|f(x) - f(1)| \geq \varepsilon_0$.
Choose $\varepsilon_0 = 1 > 0$. Let arbitrary $\delta > 0$ be given.
Choose $x = 1 - \delta/2$. Then $|x - 1| = |-\delta/2| = \delta/2 < \delta$.
Moreover, since $x = 1 - \delta/2 < 1$, we use the left branch $f(x) = 3x - 5 < 3(1) - 5 = -2$.
Since $f(1) = 1$, we have:
$|f(x) - f(1)| = |f(x) - 1| = 1 - f(x) > 1 - (-2) = 3 > 1 = \varepsilon_0$.
Therefore, f is not continuous at $x = 1$." ▪

WHY M = MAX{1, 2/E} IS MANDATORY
In piecewise functions, if you only write $M = 2/\varepsilon$, when $\varepsilon = 10$ you get $M = 0.2$, which falls into the WRONG branch ($x < 1$)! Including $\max\{1, \ldots\}$ guarantees $x > 1$!

DISCONTINUITY VIA SEQUENCES
If the question does NOT force the ε-δ definition, you can also pick sequence $x_n = 1 - 1/n \to 1$ where $f(x_n) = 3(1-1/n)-5 \to -2 \neq f(1) = 1$!

WORKED EXAMPLE 2 Riemann Sums & Rational/Irrational Non-Integrability (2025 S1 Q7)

13 MARKS EXAM Q

2025 S1 Q7 Archetype: (a) [4 marks] For $g(x) = x^2 + 1$ on $[0, 3]$ and $P_0 = \{0, 1, 2, 3\}$, compute $L(g, P_0)$ and $U(g, P_0)$.
(b) [4 marks] Let $f(x) = 4 - x^2$ if $x \in Q$, and 0 if $x \notin Q$ on $[0, 2]$. Compute $L(f, P)$ for any partition P.
(c) [5 marks] Find a positive lower bound for $\text{Upper Integral } \int_0^2 f$ and prove f is not Riemann integrable on $[0, 2]$.

PART (a) [4 MARKS] Concrete Partition Sum on $P_0 = \{0, 1, 2, 3\}$

Subintervals are $I_1=[0,1]$, $I_2=[1,2]$, $I_3=[2,3]$, each of width $\Delta x_i = 1$. Since $g(x) = x^2 + 1$ is strictly increasing on $[0, 3]$, $m_i = g(x_{i-1})$ and $M_i = g(x_i)$:
• Values: $g(0)=1$, $g(1)=2$, $g(2)=5$, $g(3)=10$.
• **Lower Sum:** $L(g, P_0) = 1(1) + 2(1) + 5(1) = \mathbf{8}$. | **Upper Sum:** $U(g, P_0) = 2(1) + 5(1) + 10(1) = \mathbf{17}$.

PART (b) [4 MARKS] Arbitrary Partition Lower Sum $L(f, P) = 0$ via Density of Irrationals

Let $P = \{x_0, x_1, \ldots, x_n\}$ be any partition of $[0, 2]$.
1. On $[0, 2]$, for $x \in Q$ we have $4 - x^2 \geq 4 - 2^2 = 0$, and for $x \notin Q$ we have $f(x) = 0$. Thus $f(x) \geq 0$ everywhere.
2. **By the density of the irrational numbers ($R \setminus Q$) in R**, every subinterval $[x_{i-1}, x_i]$ contains at least one irrational number $z_i \notin Q$, where $f(z_i) = 0$.
3. Thus $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x) = 0$ for all i, giving $L(f, P) = \sum 0 \cdot \Delta x_i = 0$!

PART (c) [5 MARKS] Bounding Upper Integral > 0 & Proving Non-Integrability

1. For any partition P of $[0, 2]$, refine it to $P' = P \cup \{1\}$ (refining decreases upper sums: $U(f, P) \geq U(f, P')$).
2. Consider the subintervals of P' lying inside $[0, 1]$. **By the density of Q in R**, every subinterval inside $[0, 1]$ contains a rational number $q \in Q$, where $f(q) = 4 - q^2 \geq 4 - 1^2 = 3$. Thus $M_i \geq 3$ on $[0, 1]$.
3. Since $M_i \geq 0$ on $[1, 2]$, we have $U(f, P) \geq U(f, P') \geq 3 \cdot (1 - 0) + 0 \cdot (2 - 1) = 3$.
4. Taking the infimum over all partitions P gives $\text{Upper Integral} \geq 3 > 0$. Since $\text{Lower Integral} = \sup_P L(f,P) = 0 \neq \text{Upper Integral}$, **$f$ is NOT Riemann integrable on $[0, 2]$**! ▪

MAGIC PHRASE FOR FULL MARKS
Whenever a function splits into $x \in Q$ and $x \notin Q$, you MUST explicitly write the phrase: **"By the density of the rationals Q (and irrationals $R \setminus Q$) in R..."**!

WHEN DOES Q VS R\Q INTEGRATE?
Only at a single point where the two branches intersect (e.g. if $4 - x^2 = 0$ at $x = 2$), the jump is 0 at that point, but over any interval of length > 0 with gap > 0, it is non-integrable!

Topic 5 · Series & Power Series: Worked Exam Guide

[!!!] ALWAYS QUESTION 8
SERIES & POWER SERIES · 14 MARKS

Comparison, AST, Ratio Test & Interval of Convergence ·

CROSS-REF SPEC §6.1–6.2

WORKED EXAMPLE 1 Infinite Series Tests & Power Series Interval (2025 S1 Q8 — 14 Marks)

14 MARKS EXAM Q

2025 S1 Q8 Archetype: (a) [6 marks] Prove whether each series converges or diverges: (i) $\sum_{n=1}^{\infty} \frac{4}{(n^3+5)}$, (ii) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+3}}$. (b) [8 marks] Find the radius of convergence R and interval of convergence I for $\sum_{n=1}^{\infty} \frac{(2^n)}{(\sqrt{n})} x^n$.

PART (a)(i) [3 MARKS] Direct Comparison Test with a Convergent p-Series

1. For all $n \geq 1$, we have $n^3 + 5 > n^3 > 0$, which implies $0 < \frac{4}{(n^3+5)} < \frac{4}{n^3}$.
2. The series $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is a **p-series with $p = 3 > 1$**, so it converges (and thus $\sum \frac{4}{n^3}$ converges).
3. Therefore, by the **Direct Comparison Test**, $\sum_{n=1}^{\infty} \frac{4}{(n^3+5)}$ **converges**! ▪

PART (a)(ii) [3 MARKS] Alternating Series Test (AST) 3-Point Checklist

Write the series as $\sum_{n=1}^{\infty} (-1)^n b_n$ where $b_n = \frac{1}{\sqrt{(n+3)}}$. We check the 3 AST hypotheses:

1. **Positivity:** $b_n = \frac{1}{\sqrt{(n+3)}} > 0$ for all $n \geq 1$.
2. **Monotone Decreasing:** Since $n+4 > n+3 > 0$, we have $\sqrt{(n+4)} > \sqrt{(n+3)} \Rightarrow b_{n+1} < b_n$ for all $n \geq 1$.
3. **Limit is Zero:** $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{\sqrt{(n+3)}} = 0$.

Therefore, by the **Alternating Series Test**, the series **converges**! ▪

PART (b) [8 MARKS] Power Series Radius R + Both Endpoints Check

- **Step 1 (Radius of Convergence R):** Here $c_n = \frac{2^n}{\sqrt{n}}$. Compute the ratio limit:
  $R = \lim_{n\to\infty} \left|\frac{c_n}{c_{n+1}}\right| = \lim_{n\to\infty} \left[\frac{(2^n)}{(\sqrt{n})} \cdot \frac{(\sqrt{(n+1)})}{(2^{n+1})}\right] = (1/2) \cdot \lim_{n\to\infty} \sqrt{(1 + 1/n)} = \mathbf{1/2}$.
  Thus the power series converges absolutely for $|x| < 1/2$ and diverges for $|x| > 1/2$.
- **Step 2 (Test Right Endpoint $x = +1/2$):** Substitute $x = 1/2$:
  $\sum_{n=1}^{\infty} \frac{(2^n)}{(\sqrt{n})}(1/2)^n = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$, which is a **$p$-series with $p = 1/2 \leq 1$ → DIVERGES!**
- **Step 3 (Test Left Endpoint $x = -1/2$):** Substitute $x = -1/2$:
  $\sum_{n=1}^{\infty} \frac{(2^n)}{(\sqrt{n})}(-1/2)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$. Since $1/\sqrt{n} > 0$ is decreasing and $\to 0$, it **CONVERGES by AST!**
- **Conclusion:** Radius $R = 1/2$, and Interval of Convergence is $I = [-1/2, 1/2)$! ▪

ENDPOINT TESTING IS HALF THE MARKS!
In an 8-mark Power Series question, finding $R = 1/2$ is worth only 4 marks; testing $x = +R$ and $x = -R$ and writing the exact bracket $[-1/2, 1/2)$ is worth the other 4 marks!

RATIO TEST ON FACTORIALS
For $a_n = \frac{2^n}{n!}$, $\left|\frac{a_{n+1}}{a_n}\right| = \left[\frac{2^{n+1}}{(n+1)!}\right] \cdot \left[\frac{n!}{2^n}\right] = \frac{2}{(n+1)} \to 0 < 1$ → Converges by Ratio Test!

WORKED EXAMPLE 2 Taylor's Theorem & Uniform Convergence (Exam Q9 — 10 Marks)

10 MARKS FINAL Q

Exam Q9 Archetype: (a) [5 marks] Let $g(x) = \cos(x) - 1$. Compute degree 3 Taylor polynomial $T_3(x)$ centered at 0, and use Taylor's Theorem to show $|g(x) - T_3(x)| \leq x^4 / 24$ for all $x \in \mathbb{R}$. (b) [5 marks] For $f_n(x) = \frac{x^2}{(n+x^2)}$ on $[0, 1]$, find the pointwise limit $f(x)$ and prove $f_n \to f$ uniformly on $[0, 1]$.

PART (a) [5 MARKS] Taylor Polynomial $T_3(x)$ & Lagrange Remainder Bound

1. **Compute Derivatives at $a = 0$:**
   $g(x) = \cos(x) - 1 \Rightarrow g(0) = 0$; $g'(x) = -\sin(x) \Rightarrow g'(0) = 0$; $g''(x) = -\cos(x) \Rightarrow g''(0) = -1$; $g'''(x) = \sin(x) \Rightarrow g'''(0) = 0$; $g^{(4)}(x) = \cos(x)$.
2. **Assemble $T_3(x)$:** $T_3(x) = g(0) + g'(0)x + [g''(0)/2!]x^2 + [g'''(0)/3!]x^3 = \mathbf{-x^2/2}$.
3. **Apply Taylor's Theorem (Lagrange Remainder for $k = 3$):**
   For any $x \in \mathbb{R}$, by Taylor's Theorem there exists c between 0 and x such that:
   $g(x) - T_3(x) = R_3(x) = [g^{(4)}(c) / 4!] x^4 = [\cos(c) / 24] x^4$.
   Taking absolute values and using $|\cos(c)| \leq 1$ for all $c \in \mathbb{R}$:
   $|g(x) - T_3(x)| = |\cos(c)| \cdot |x|^4 / 24 \leq 1 \cdot x^4/24 = x^4/24$. ▪

PART (b) [5 MARKS] Pointwise Limit & Uniform Convergence via Supremum Norm M_n

1. **Pointwise Limit $f(x)$:** Fix any $x \in [0, 1]$. Since x^2 is constant with respect to n and $n + x^2 \to \infty$ as $n \to \infty$, we have:
   $f(x) = \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{x^2}{(n+x^2)} = 0$. Thus $f(x) = 0$ for all $x \in [0, 1]$.
2. **Uniform Convergence Bound (Eliminate x uniformly!):**
   For every $x \in [0, 1]$, the numerator satisfies $0 \leq x^2 \leq 1$, and the denominator satisfies $n + x^2 \geq n > 0$. Therefore:
   $|f_n(x) - f(x)| = \left| \frac{x^2}{(n+x^2)} - 0 \right| = \frac{x^2}{(n+x^2)} \leq 1/n$ for all $x \in [0, 1]$.
3. Taking the supremum over $x \in [0, 1]$ gives:
   $0 \leq M_n = \sup_{x \in [0,1]} |f_n(x) - f(x)| \leq 1/n$.
   Since $\lim_{n\to\infty} (1/n) = 0$, by the Squeeze Theorem $\lim_{n\to\infty} M_n = 0$.
   Hence **(f_n) converges uniformly to $f = 0$ on $[0, 1]$**! ▪

HOW TO DISPROVE UNIFORM CONVERGENCE
To show $f_n(x) = x^n$ on $[0, 1]$ is **NOT** uniformly convergent: each f_n is continuous, but the pointwise limit $f(x) = 0$ (on $[0,1)$) and 1 (at $x=1$) is discontinuous! (Uniform limit of continuous functions must be continuous).

TAYLOR DEGREE K REMAINDER INDEX
For degree k polynomial $T_k(x)$, the Lagrange remainder ALWAYS uses the **$(k+1)$-th derivative** divided by $(k+1)!$: for $T_3(x)$, use $g^{(4)}(c)/4!$!